The section on one side is the emitter and the section on the opposite side is the collector. The middle section is called the base and forms two junctions between the emitter and collector. The above fig 10.8 (i) & (ii) shows the symbols of n-p-n & p-n-p transistors.

Note that emitter is shown by an arrow which indicates the direction of conventional current flow with forward bias. For n-p-n connection, it is clear that conventional current flow out of the emitter as indicated by the outgoing arrow in fig 10.8(i). Similarly, for p-n-p connection, the conventional current flows into the emitter as indicated by inward arrow in fig 10.8(ii).

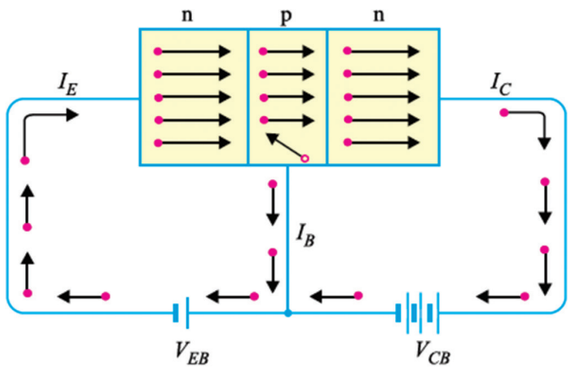
Transistor Action

The emitter base junction of a transistor is forward biased where as collector base junction is reverse biased. If for a moment, we ignore the presence of emitter base junction, then practically no current would flow in the collector circuit because of the reverse bias. However, if the emitter base junction is also present, then forward bias on it causes the emitter current to flow. It is seen that this emitter current almost entirely flows in the collector circuit. Therefore, the current in the collector circuit depends upon the emitter current. If the emitter current is zero, then collector current is nearly zero. However if the emitter current is 1ma, then collector current is also about 1ma. This is precisely what happens in a transistor.

We shall now discuss the transistor action for npn and pnp transistors.

10.5(i) Working of npn – Transistor

Fig 10.9 shows the npn-transistor with forward bias to emitter base junction and reverse bias to collector base junction. The forward bias causes the electrons in the n-type emitter to flow towards the base. This constitutes the emitter current I_E. As these electrons flow through the p-type base, they tend to combine with holes. The base is lightly doped and very thin. Therefore, only a few electrons less than 5% combine with holes to constitute base current I_B. The remainder more than 95% cross over into the collector region to constitute collector current I_C. In this way, almost the entire emitter current flows in the collector circuit. It is clear that emitter current is the sum of collector and base currents. ie, $I_E=I_B+I_C$.



Basic Connection of *npn* Transistor

Fig 10.9 *n-p-n Transistor*

10.5(ii) Working of pnp - Transistor

Fig 10.10 shows the basic connection of a pnp transistor. The forward bias causes the holes in the p-type emitter to flow towards the base. This constitutes the emitter current I_E. As these holes cross into n-type base, they tend to combine with the electrons. As the base is lightly doped and very thin. Therefore, only a few

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